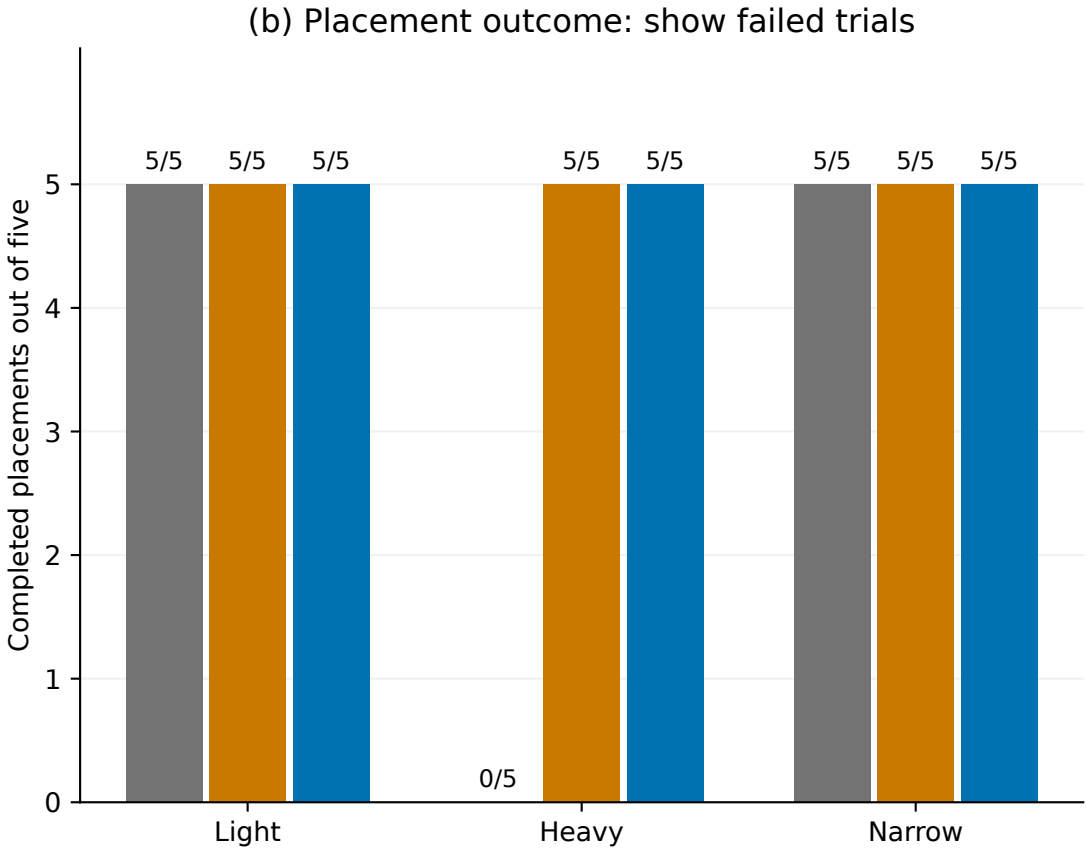
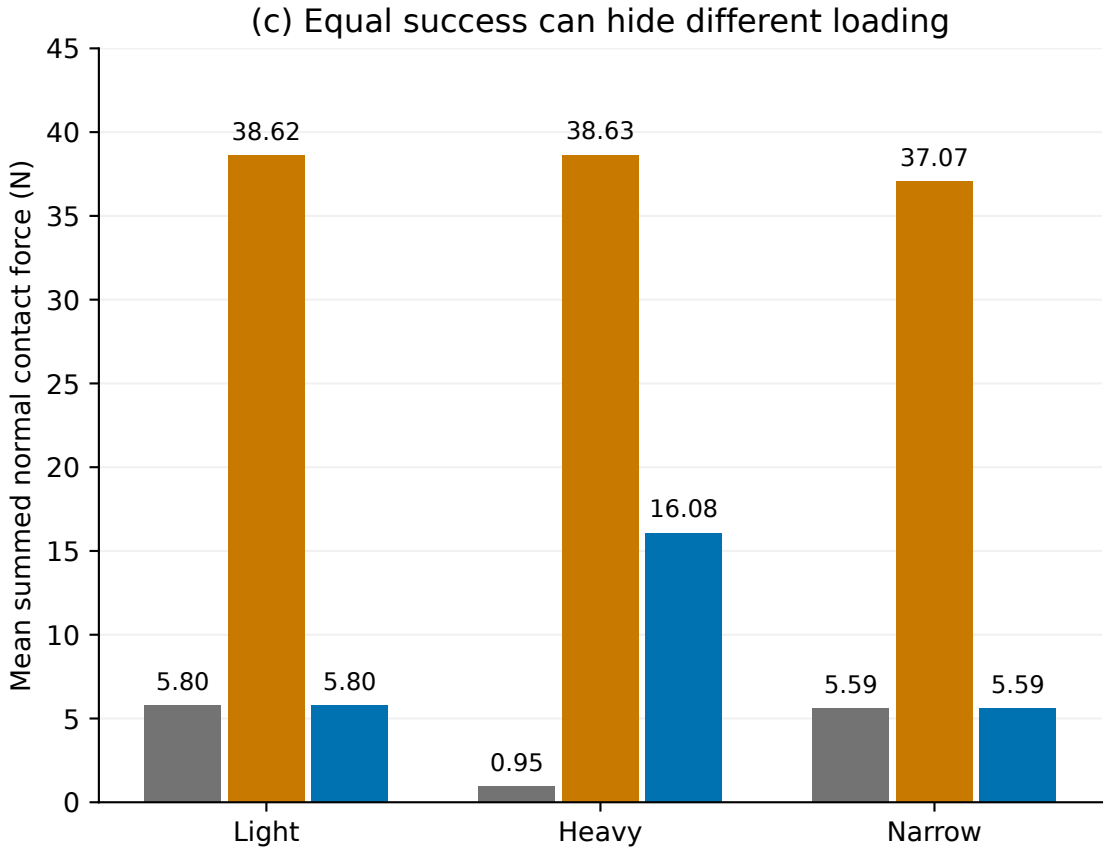
Why similar movements are not equivalent handling outcomes

Same commanded arm trajectory; recorded simulation results, no new trials

(a) Same light/heavy geometry; mass and friction both change

Profile	Object dimensions	Mass	Friction coefficient
Light	50 × 50 × 60 mm	0.08 kg	0.8
Heavy	50 × 50 × 60 mm	0.50 kg	0.4
Narrow	24 × 24 × 80 mm	0.12 kg	0.5





Motor limit is per finger; contact force sums both fingers

Fixed low (3 N) Fixed high (20 N) Slip feedback (3–20 N)